



Active micro-posts control the morphology of underlying compliant substrates



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Impact statement

Cilia are microscale flexible filaments that are anchored to the soft tissue in, for example, the lungs, inner ears, and stomach and perform functions that are essential to our viability. Theoretical and experimental studies of biological and synthetic cilia typically do not consider the elasticity of the underlying substrate that anchors the cilia, nor the energy transduction from chemical reactions that fuels their mutual interactions. Hence, we lack detailed knowledge about the chemical and mechanical cues that enable communication between the cilia, substrate, and surrounding fluid flowing in the body. To gain further insight into forces affecting the ciliary behavior, we developed a model that encompasses the elasticity of the substrate, the flexibility of the cilia, and the enzymatic reactions that contribute to the fluid motion. Our results reveal how chemically driven fluid motion affects the shape of both the bendable cilia and deformable soft substrate, and in turn how the deformed surfaces alter the flow behavior and thus trigger continuous feedback among chemistry, the fluid and the surfaces. Understanding the latter feedback is critical to unraveling the cilia's biological behavior and fabricating synthetic analogues that expand the functionality of fluidic devices.

In fluid-filled chambers, the motion of chemically active cilia tethered to soft substrates can deform the underlying layer and thus modify the delicate interplay between chemistry and mechanics in the system. To determine the effect of chemo-mechanical interactions, we developed computational models to simulate the system's dynamics when chemical reactions at enzyme-coated cilia and substrates spontaneously generate fluid vortices. The flowing fluids, in turn, morph the ciliated layers into controllable shapes, including “scrolls,” “envelopes,” and convex “bridges,” depending on the competition between the system's chemical reactivity and mechanical properties. Through this model, we designed a ciliated substrate that spontaneously and repeatedly rolls into a “scroll” and unrolls into a flat surface, showing a significant level of autonomous behavior. Our findings yield design rules for facilitating specific applications of fluidic devices and highlight the importance of interactions between mobile cilia and the underlying soft layer in both biological and synthetic systems.

Introduction

Cilia are flexible, microscale “hairs” that are anchored onto various compliant surfaces in living organisms. The motion of these cilia in the surrounding fluid plays a crucial role in, for example, clearing lungs of harmful particulates, sending messages from the ear to the brain, and transporting cells in the reproductive system. On the skin of marine organisms, the beating of cilia enables not only the transport of food to the animal's mouth, but also drives its movement. The surfaces of cilia are soft, as are the surfaces of the lungs, inner ears, cells, skin, and other biomaterials underneath the ciliated layer. In the presence of flowing fluids, the ciliated system (cilia plus deformable substrate) is replete with complex, interlocking fluid-structure interactions. The fluid flow exerts a force on the

compliant cilia and soft substrate (see **Figure 1**). The fluid-driven motion and deformation of the cilia also affects the shape of this soft substrate. The deformed surfaces, in turn, exert opposing forces, which alter the flow behavior, and thus trigger continuous feedback between the fluid and the surfaces. In most computational models, however, the substrate is assumed to be a flat, nondeformable layer. To date, there are few experimental^{1–4} or modeling studies^{5,6} that characterize the interconnection among the fluid flow (hydrodynamics), motion of the bendable cilia, and shape changes of the underlying soft substrate. Understanding the latter interactions is vital to developing a more complete understanding of the behavior of biological cilia, as well as designing ciliated surfaces to extend the functionality of fluidic devices.

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Herein, we introduce a computational approach to simulate the simultaneous dynamic interactions among elastic cilia, the underlying flexible substrate, and the flowing fluid. Our simulations include the key interchange between chemistry and mechanics (chemo-mechanical transduction) that induces the initial motion of the surrounding fluid. Using this approach, we focus on a self-contained system where a flexible ciliated layer is enclosed in fluid-filled chamber. As shown here, the fluid-driven motion of chemically active cilia alters the shape and curvature of the underlying soft substrate. For example, depending on the cilia's length, a flat ciliated layer can spontaneously morph into a scroll, a folded envelope, or bridge-shaped object. We also demonstrate how the chemical specificity of the reactions affect the transformation of the compliant systems. The layer's final steady-state structure is regulated by the chemical reactivity of the underlying layer and that of the grafted cilia, the mechanical properties of these flexible structures, and steric interactions between the cilia. Additionally, the dynamic structural changes in the system modify the velocity of the enclosed fluid, which in turn affects the subsequent morphologies of the entire "carpet" (cilia plus the substrate). The study reveals the importance of contributions from the soft substrate in determining the complex behavior of the entire system.

Our approach builds on our recent studies of chemo-mechanical transduction in flexible, two-dimensional (2D) sheets coated with catalysts and immersed in a fluid-filled chamber. In the presence of the appropriate reagents, the catalysts convert reactants into products that occupy different volumes. The latter differences in volume give rise to density gradients, which generate a force that acts on the fluid and drives it to move. The latter process is referred to as the solutal buoyancy mechanism. The driven fluid, in turns, alters the shape of the compliant sheet through fluid-structure interactions.^{7–11} The chemo-mechanical transduction occurs as the energy released from the catalytic reactions propels the fluid and drives it to perform mechanical work.

Here, the flexible, 2D layers function as an underlying substrate for an array of grafted micro-posts that represent the cilia. In all the examples, the posts are taken to be catalyst-coated and thus chemically active. The sheet, on the other hand, can also be catalyst-coated and thereby chemically active or bare of catalysts and thus chemically passive. The model allows us to isolate the mechanical strain induced in the sheet by the active cilia, and thus obtain insight into mechanical signaling mechanisms that enable separated cilia to communicate and coordinate their motion, as well as uncover mechanical signaling between the cilia and the substrate that in turn affects ciliary motion. Coating the sheets with chemically distinct catalysts provides a means to determine how different classes of chemical reactions affect chemo-mechanical communications and hence, controllably modify the system's functionality.

Materials and methods

The fluid-filled micro-chamber contains an elastic sheet, and 24 micro-posts anchored onto the surface of the sheet (Figure 1a). In the simulations, the chemical pump is formed by setting a specified number of sheet and/or micro-posts nodes as catalytic sites. We consider both scenarios where the nodes of the sheet remain uncoated ("passive;" colored in green), or where the nodes of the sheet (or micro-posts) are coated with different enzymes ("active"). The active posts are marked in blue for CAT (catalase), in pink for AP (acid phosphatase), and brown for UR (urease) (Figure 1b). The chemical composition of the solution is controlled by reactions with the immobilized enzymes, which transform reactants to products. The products can occupy different volumes than the reactants, leading to local density variations in the fluid that give rise to a buoyancy force per unit volume, $\mathbf{F}_b = \mathbf{g} \rho_0 \sum \beta_i C_i$, which acts on the fluid. Here, ρ_0 is the density of the solvent, and $\mathbf{g} = -g\hat{\mathbf{z}}$ ($\hat{\mathbf{z}}$ is the unit vector in the positive z -axis direction) is the acceleration vector due to gravity. Additionally, C_i is the concentration of chemical species i , and $\beta_i = 1/\rho_0(\partial \rho / \partial C_i)$ is the corresponding solutal expansion coefficient. The solutal buoyancy force drives the spontaneous motion of the fluid, which in turn imposes a fluid drag on objects immersed in the solution. Figure 1c shows the side view of the flow field generated by a CAT-coated sheet located at the bottom of a micro-chamber. With the addition of hydrogen peroxide, the catalytic reaction decomposes this reactant into less dense products (i.e., water and oxygen). The color bar depicts the concentration of hydrogen peroxide in the solution after one hour. The fluid motion toward the sheet along the bottom surface is termed "inward" flow.

Both the active and passive nodes of the compliant structures are described as spherical beads of radius $a \sim 1.3 \Delta x$ (Δx is the mesh size) that are arranged into a mesh, with r_k representing the position of the respective node. The nodes are interlinked by elastic bonds (springs). The forces acting on the bead nodes are given by $\mathbf{F}^{\text{bead}} = \mathbf{F}^e + \mathbf{F}^s + \mathbf{F}^g$, where the last three terms represent the respective elastic, steric, and gravitation forces. The elastic forces $\mathbf{F}^e$ (acting as the elastic inter-bead bonds), are characterized by the stretching (κ_s) and bending moduli (κ_b) and are governed by the linear constitutive relations for a Kirchhoff rod^{12,13} (see Supplementary information [SI] Appendix). The steric forces on the bead, $\mathbf{F}^s$, constitute the sum of the repulsive interactions between two nodes, $\mathbf{F}^{nn}$, and between a node and any of the nodes on six confining walls, $\mathbf{F}^{nw}$. The steric repulsion force, $\mathbf{F}^s = -\nabla U(r)$, is computed from gradients of a Morse potential, which ensures that the equilibrium node-node (nn) distance and the node-wall (nw) distance is equal to r_0 (see SI Appendix). The gravitational force acting on each bead is given by $\mathbf{F}^g = V(\rho_b - \rho_0)\mathbf{g}$, where V is the effective volume of each node, and the density of the bead (ρ_b) is assumed to be greater than the density of the solvent (ρ_0). To conserve the

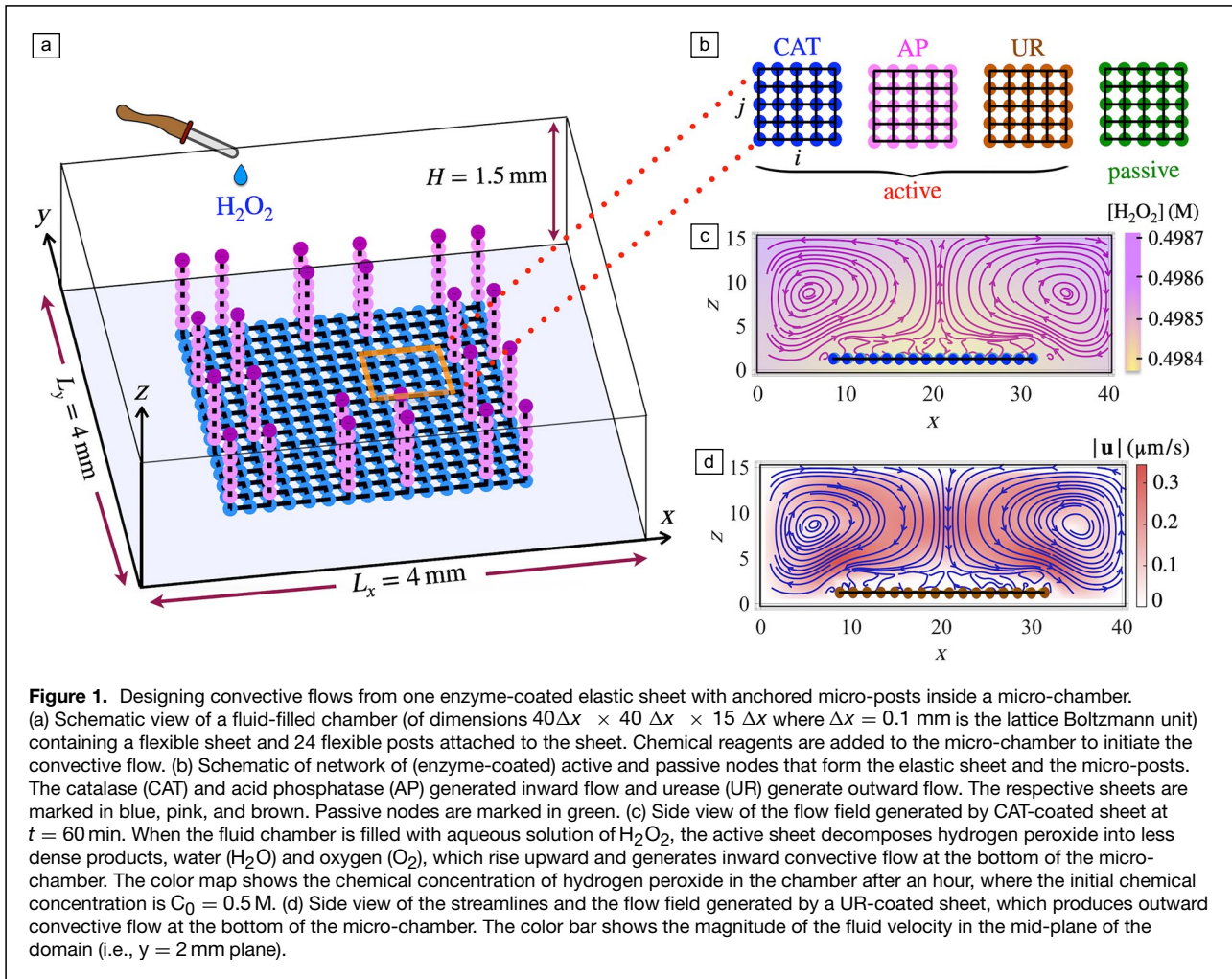


Figure 1. Designing convective flows from one enzyme-coated elastic sheet with anchored micro-posts inside a micro-chamber. (a) Schematic view of a fluid-filled chamber (of dimensions $40\Delta x \times 40\Delta x \times 15\Delta x$ where $\Delta x = 0.1$ mm is the lattice Boltzmann unit) containing a flexible sheet and 24 flexible posts attached to the sheet. Chemical reagents are added to the micro-chamber to initiate the convective flow. (b) Schematic of network of (enzyme-coated) active and passive nodes that form the elastic sheet and the micro-posts. The catalase (CAT) and acid phosphatase (AP) generated inward flow and urease (UR) generate outward flow. The respective sheets are marked in blue, pink, and brown. Passive nodes are marked in green. (c) Side view of the flow field generated by CAT-coated sheet at $t = 60$ min. When the fluid chamber is filled with aqueous solution of H_2O_2 , the active sheet decomposes hydrogen peroxide into less dense products, water (H_2O) and oxygen (O_2), which rise upward and generates inward convective flow at the bottom of the micro-chamber. The color map shows the chemical concentration of hydrogen peroxide in the chamber after an hour, where the initial chemical concentration is $C_0 = 0.5$ M. (d) Side view of the streamlines and the flow field generated by a UR-coated sheet, which produces outward convective flow at the bottom of the micro-chamber. The color bar shows the magnitude of the fluid velocity in the mid-plane of the domain (i.e., $y = 2$ mm plane).

momentum exchange between the structure and the fluid, the forces acting on each bead are balanced by the hydrodynamic drag force $\mathbf{F}^h = -(\mathbf{F}^{\text{bead}})$.

The dynamic interactions between the dissolved chemicals and the elastic structure (sheet and the attached posts) are described by the following coupled equations: the continuity and Navier–Stokes (in the Boussinesq approximation¹⁴) equations for an incompressible flow; the equation for the advection, diffusion, and reaction of the dissolved chemical species C_i ; and the equation for the velocity of the bead nodes:

$$\nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho_0} \nabla p + \nu \nabla^2 \mathbf{u} + \frac{1}{\rho_0} (\mathbf{F}_b + \mathbf{F}^{\text{IB}}), \quad (2)$$

$$\frac{\partial C_i}{\partial t} + (\mathbf{u} \cdot \nabla) C_i = D_i \nabla^2 C_i \pm SK_d \sum_{k=1}^M \delta(\mathbf{r}_k), \quad (3)$$

$$\frac{\partial \mathbf{r}_k}{\partial t} = \mathbf{u}(\mathbf{r}_k). \quad (4)$$

Here, $\mathbf{u}$ is the fluid velocity, p is the fluid pressure, and ν is the kinematic viscosity. The body forces acting on the fluid have contributions from the solutal buoyancy force, $\mathbf{F}_b$, and forces exerted on the fluid by the beads, $\mathbf{F}^{\text{IB}} = -\sum \mathbf{F}^h$, which are calculated *via* the immersed boundary method¹⁵ (IBM), providing fluid–structure interactions.

The chemicals are consumed or produced at the position of the enzyme-coated bead $\mathbf{r}_k$, which make up the posts that have a reaction rate given by SK_d , where S is the area coated by catalyst. The “ $\pm$ ” sign in Equation 3 represents either production or consumption of the reactants and products. We assume that the catalytic reaction on the sheet and posts follows the Michaelis–Menten kinetics,¹⁶ where the rate of consumption per unit area of the catalytic coverage is given by:

$$K_d = \frac{r_{m,\text{post}}^e C_i}{K_M + C_i}. \quad (5)$$



Here, $r_{\text{m,post}}^e = k_e[E]$ (in units of $\text{mol m}^{-2} \text{s}^{-1}$) incorporates the maximum reaction rate per molecule of enzyme, k_e , with areal enzyme concentration $[E]$, and K_M (in units of molarity, M) is the Michaelis constant. Both reactants and products undergo diffusion within the solution and are advected by the generated fluid flow. In our simulations, we use reaction rates r_{m}^e that are comparable to experimental values reported by Sengupta et al.¹⁷ and Maiti et al.¹⁸ Supplementary Tables SI and SII summarize chemical parameters and reaction rates used in our simulations.

We use no-slip boundary conditions for the fluid flow ($\mathbf{u} = 0$) at the walls of the micro-chamber, and no-flux conditions, $\mathbf{n} \cdot \nabla C_i = 0$, for chemical concentration C_i at the confining walls ($\mathbf{n}$ is the surface normal pointing into the fluid domain) with initial conditions $C_i(t = 0) = C_{0,i}$. The set of governing equations (1–4), along with these boundary conditions for the fluid velocity and chemical concentrations, are solved numerically. The continuity and Navier–Stokes equations are solved using the lattice Boltzmann method¹⁹ (LBM) with a single relaxation time D3Q19 scheme.²⁰ A finite difference approach with a forward-time central-space (FTCS) scheme is used to solve the advection-reaction-diffusion of chemicals. The immersed boundary (IBM) approach is used to capture the fluid–structure interactions between the elastic posts and fluid. In the IBM model, each node of elastic post is modeled by a sphere with effective hydrodynamics radius $a \sim 1.3 \Delta x$ that experiences fluid drag characterized by the mobility $M = (6\pi\eta a)^{-1}$. The forces F^e exerted by the nodes of the elastic structure on the fluid, which are calculated using the IBM method, provide zero fluid velocities at the discretization nodes of the elastic structure. Therefore, the IBM approach approximates no-slip conditions for the fluid velocities at the boundary nodes, as well as no fluid permeation through the nodes of the elastic structure.

Values for bending stiffness (κ_b) used to parametrize the elastic properties of the sheets and posts in our simulations are extracted from experimental values of flexural rigidity B , using a procedure detailed in the Supplementary information (see also Manna et al.⁹). In short, we use a one-dimensional differential equation to analytically describe a linear beam clamped at one end (for the tethered posts) and subjected to a given transversal force F at the free end. We then simulate a discretized version of the post with the same length, under the action of the same force F , and adjust the value of κ_b to match the transversal displacement of the nodes that is obtained by solving the 1D differential equation. The procedure provides a means to relate the simulation parameters κ_b to experimentally used values of flexural rigidities, B , as described in the SI Appendix.^{21–23} The values for κ_s were chosen to ensure that the posts do not stretch more than 2% under the action of the applied fluid flows. The resulting bending and stretching moduli for the sheet and posts were taken $\kappa_b = 0.22 \text{ pN mm}^2$ and $\kappa_s = 20 \text{ pN}$, respectively.

The size of computational domain is $40\Delta x \times 40\Delta x \times 15\Delta x$ where the lattice Boltzmann unit Δx is $100 \mu\text{m}$ (see Figure 1a). In the elastic structure, which includes the sheet and attached micro-posts, distance between two neighboring nodes (beads) is set to $1.5 \Delta x$. The lateral dimensions of elastic sheet are $22.5 \Delta x \times 22.5 \Delta x$ with thickness of $2.6 \Delta x$. The time step of simulations is $\Delta t = 1.67 \times 10^{-3} \text{ s}$.

Physical parameters that control chemistry-driven, fluid-mediated transformations of the elastic structures can be combined into several dimensionless numbers that describe scaling properties and characterize behavior of the system. The ratio of solutal to viscous forces is expressed by the dimensionless Grashof number, $\text{Gr} = \frac{g\beta C_0 L^3}{\nu^2}$, where β is the solutal expansion coefficient, C_0 is the initial concentration of chemical, ν is the kinematic viscosity of the fluid, and L is the characteristic size of the chamber. The typical Grashof number in our system is about 10^2 . Relevant Reynolds ($\text{Re} = \frac{UL}{\nu}$) and Peclet ($\text{Pe} = \frac{UL}{D}$) numbers are on the order of 10^{-3} and 1, respectively (D is the diffusion coefficient, U is the typical fluid velocity in the domain).

The elastic properties of the sheet are characterized by the dimensionless flexural rigidity, which is the ratio of bending stress in the sheet to viscous stress in the fluid. The typical dimensionless elastoviscous number for a sheet of bending rigidity $B' = 10^{-16} \text{ Nm}$ (see SI Appendix) is about ~ 0.1 . In our system, the typical ratio of cilia length to domain length is $1/6$. During structural deformations, the elastic forces associated with the bending compete with the solutal buoyancy force exerted on the fluid. The corresponding dimensionless ratio, $\frac{B'/ds}{\rho_0 g \beta C L^2 t} \sim 10^{-2}$, is quite small indicating that our structures are easily bent by chemically driven convective flows. Here, ds is the distance between two neighboring nodes, $L = 2.25 \text{ mm}$ is the sheet lateral size and $t = 0.26 \text{ mm}$ is the sheet thickness. We take the density difference between the material of sheet and solution $\rho_s - \rho_0 = 0.035 \text{ Kg m}^{-3}$, corresponding to a gravitational force $F^g = (\rho_s - \rho_0)gL^2 t = 4.5 \times 10^{-10} \text{ N}$ acting on the sheet. The comparison between gravity and bending is measured by elastogravitational number, $\frac{F^g}{B'/ds}$, which is of order 10^2 in our simulations. There exist both synthetic and biological thin films that exhibit a flexural rigidity comparable to the material envisioned in our model. For example, ultrathin flexible sheets of functionalized nanoparticles^{21,22} (NPs), photocross-linkable polymer films,^{24,25} and layers composed of oleosin surfactant proteins²³ can be utilized to form chemically active compliant materials by incorporating appropriate catalysts²⁶ into these films. These materials could enable experimental fabrication of systems realized in our simulations.

Results and discussion

Before focusing on each case in detail, we note that these examples all illustrate a common theme: the system's final structure is determined by a competition between mechanisms that promote chemical reactivity and those that inhibit

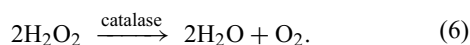


mechanical deformation. Notably, increasing the length of the posts increases the number of active beads in the solution and thereby increases the effects due to chemical reactivity (and gravity). In particular, the reaction rate in the solution is proportional to SK_d , where S is the area coated by the catalyst and K_d is the Michaelis–Menten rate constant (Equation 5). Increasing the post length increases its active surface area (S) and thus increases the total reaction rate. Simultaneously, increases in the reaction rate give rise to increases in the velocity of the flowing fluid, $\mathbf{u}$. Due to inherent feedback, the increase in $\mathbf{u}$, in turn, accelerates the rate of reaction by transporting the reactant-abundant solution to the catalyst at a rate faster than could be achieved just through diffusion. The latter effects promote chemical reactivity (i.e., the ability of individual molecules to take part in the chemical reactions in the system) and hence promote chemo-mechanical transduction in these active systems.

Conversely, the increase in fluid velocity, $\mathbf{u}$, also increases the drag forces imposed on the micro-posts through fluid–structure interactions. Resistance of the structure to these forces now involves a greater cost in energy to stretch and bend the elastic sheet. Hence, these forces hinder the system’s mechanical response to the chemical stimuli. This fundamental competition plays a critical role in structure formation.

The chemically compliant substrate is modeled as a square sheet that is 2.25 mm $\times$ 2.25 mm in the lateral directions and is 0.26 mm in width. The sheet is composed of beads shown in blue in Figure 1a and interconnected with elastic bonds shown with black lines. The 24 anchored micro-posts, which also consist of elastically connected beads, are placed symmetrically with respect to the perpendicular plane that cuts through the center of the sheet. Note that the system exhibits twofold symmetry; along the y -direction, four posts in a row are spaced uniformly, but along the x -direction, six posts in a row are separated as shown in Figure 1a. The symmetry of the structure prevents the sheet from undergoing translational motion. The initially flat sheet with the anchored micro-posts is immersed in a fluid-filled microchannel, parallel to the bottom wall (see Figure 1). We consider three representative lengths for the micro-posts ($L = 0.45$ mm, $L = 0.6$ mm, and $L = 0.75$ mm), while maintaining the bending and stretching moduli for both the sheet and micro-posts at $\kappa_b = 0.22$ pN mm² and $\kappa_s = 20$ pN, respectively.

The active beads in the system are shown in Figure 1b with colors that indicate one of the following enzymes: catalase (CAT), acid phosphatase (AP), and urease (UR). Adding hydrogen peroxide (H_2O_2) to the solution triggers the CAT-coated sites to decompose hydrogen peroxide into water (H_2O) and oxygen (O_2):



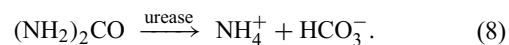
Similarly, adding p-nitrophenylphosphate hexahydrate (PNPP) to the solution triggers AP to decompose this

reactant into p-nitrophenol (PNP) and monosodium phosphate (NaH_2PO_4):



In both of the previously discussed cases, products of the reactions are less dense than reactants, causing the product-rich fluid near the active nodes to rise upward to the top wall. Due to the continuity of the fluid in the closed micro-chamber, the flow near the top splits into two equal streams. Each stream then flows downward along the nearest sidewall and back toward the center of the sheet, forming “inward” flow along the bottom surface of the micro-chamber (see Figure 1c).

The case of UR-coated nodes is different from the cases involving CAT and AP; by introducing urea ($(\text{NH}_2)_2\text{CO}$) into the aqueous solution, the enzyme urease decomposes urea into *more dense* products ammonium bicarbonate (NH_4^+ and HCO_3^- ions):



In this case, the denser products near the active nodes generate “outward” flow as they initially move along the bottom surface of the micro-chamber, away from the enzyme-coated parts, and then circulate to return to the bottom wall (Figure 1d).

With this choice of reagents, we investigate synergistic interactions that emerge when (1) the posts are active and the surface of the layer is passive; (2) active posts and active sheet both induce inward flow; and (3) the sheets generate inward flow, but the posts produce outward flow. For each of the cases, we analyze the effect of the post length on the formation of the final structure. To facilitate comparisons among the different cases, **Figure 2** displays the final steady-state morphology for each of these different scenarios. In Figures 3, 4, and 5, the images in row a show shapes of the compliant structures; row b depicts the flow lines generated by the reaction in the middle section of the domain; row c shows the contour plots for the (average) mean curvature of the sheet, $H = (\kappa_1 + \kappa_2)/2$, where κ_1 and κ_2 are the two principal axes of curvatures for the sheet; row d displays contour plots for the areal strains within 2D flexible layer. Areal strain is defined as the change in the area of a unit square at the final configuration relative to its initial undeformed value (see panel b in Figure 1). Panel e (in Figures 3–5) illustrate the time-dependent behavior of the mean absolute fluid velocity (in the domain), which characterizes drag forces imposed by the flow on the elastic structure, and panel f (in Figures 3–5) show the temporal evolution of the mean curvature $H = (\kappa_1 + \kappa_2)/2$ of the sheet. As we discuss below, varying the length of the posts allows us to compare forces due to gravity, chemical reactivity, and steric repulsion.

Shape morphing of an elastic passive sheet with AP-coated micro-posts

To illustrate how chemical activity at the micro-posts affects the dynamics of a passive substrate, we examine AP-coated

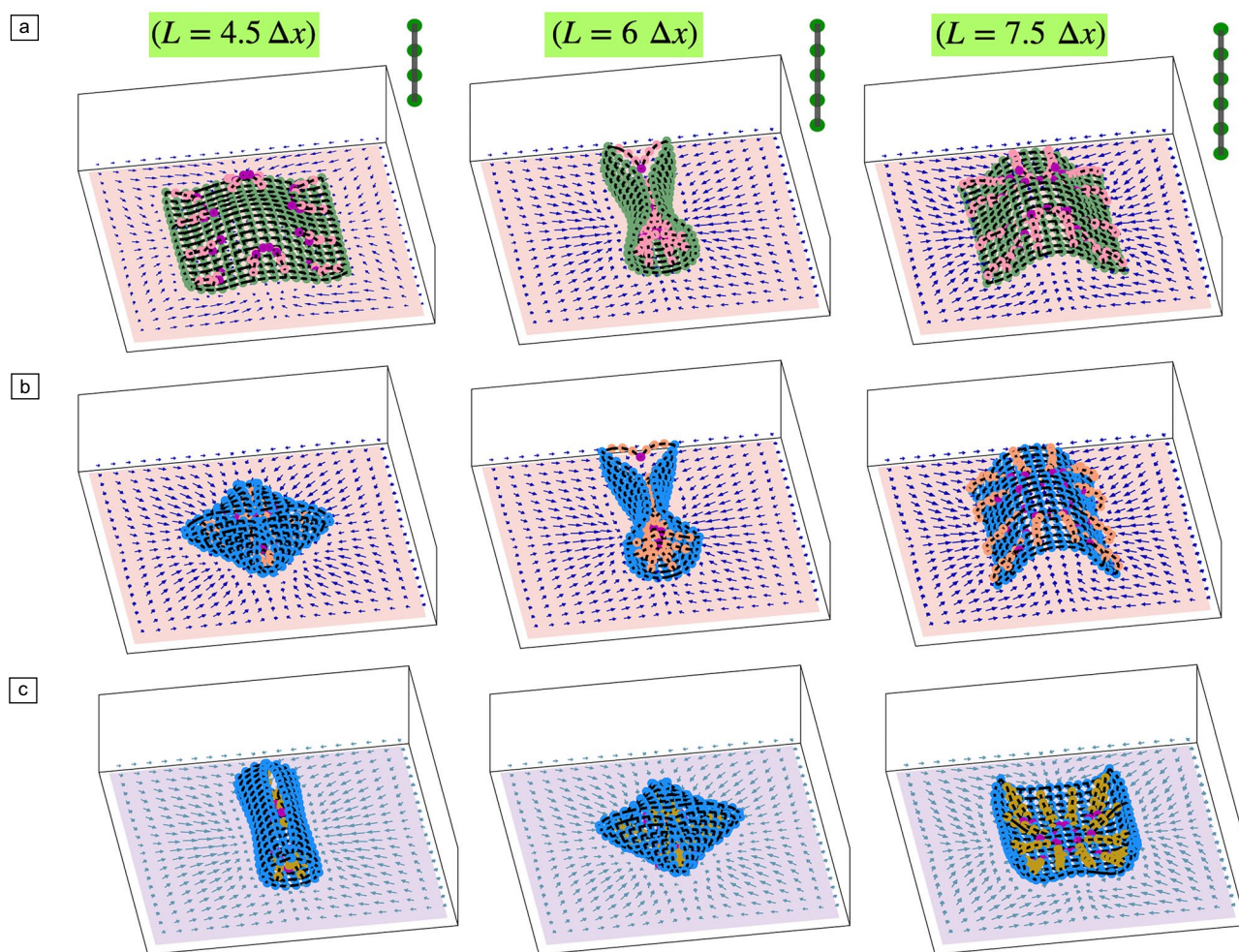


Figure 2. Multiple shape changes of an elastic (passive/active) sheet with anchored active micro-posts of different lengths. Final configuration of an elastic (a) *passive* sheet (colored in green) anchored with AP-coated micro-posts of different lengths (colored in pink) with $L = 4.5 \Delta x$, $L = 6 \Delta x$, and $L = 7.5 \Delta x$, where $\Delta x = 0.1$ mm. With the introduction of an aqueous solution of PNPP, the inward fluid flow due to catalytic reaction at the AP-coated posts transform the elastic sheet into 3D-folded structures (see Movie S1); (b) active CAT-coated sheet (colored in blue) with AP-coated micro-posts (colored in pink). In the presence of an aqueous solution of H_2O_2 and PNPP, the combination of upward fluid flow generated by the catalytic reactions on the CAT-coated sheet and AP-coated posts gives rise to these folded structures for different lengths of the micro-posts (see Movie S2); (c) active CAT-coated sheet (colored with blue) anchored with UR-coated micro-posts (colored with brown) with different lengths. With the introduction of H_2O_2 and urea, the combination of opposing inward and outward flows generated by CAT-coated sheet and UR-coated posts generates different sheet structures for different lengths of the micro-posts (see Movie S3). The tips of the micro-posts are colored in magenta. The configurations of elastic sheets are after 550 minutes from the start of catalytic reaction. The bending and stretching moduli of the sheet are $\kappa_b = 0.22$ pN mm² and $\kappa_s = 20$ pN, respectively. Reaction rates are $r_{m, post}^{AP} = 4.7 \times 10^{-8}$ mol m⁻² s⁻¹ (panels a and b), $r_{m, sheet}^{CAT} = 4 \times 10^{-7}$ mol m⁻² s⁻¹ (panel b), $r_{m, sheet}^{CAT} = 3.6 \times 10^{-6}$ mol m⁻² s⁻¹ (panel c), and $r_{m, post}^{UR} = 1.8 \times 10^{-6}$ mol m⁻² s⁻¹ (panel c).

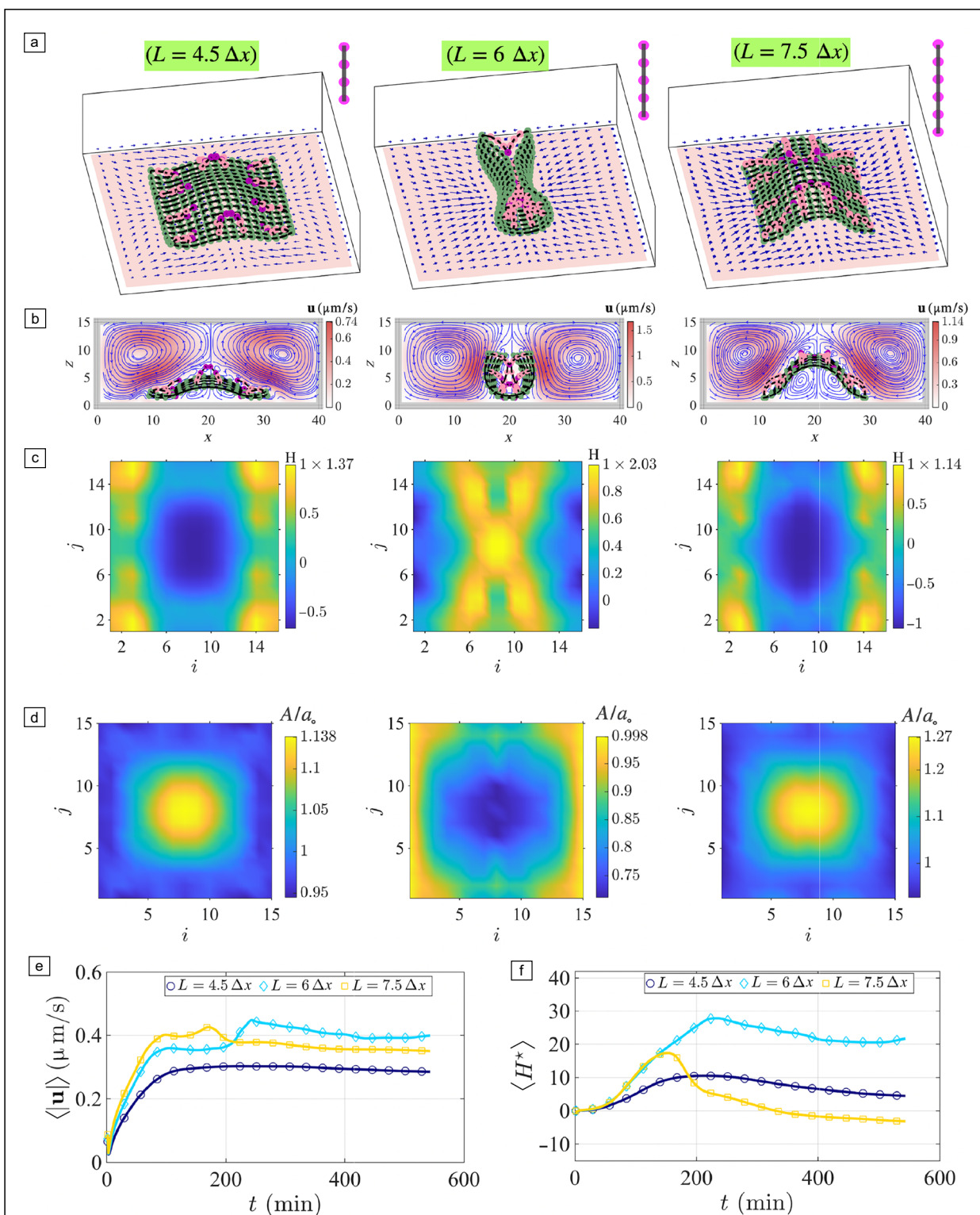
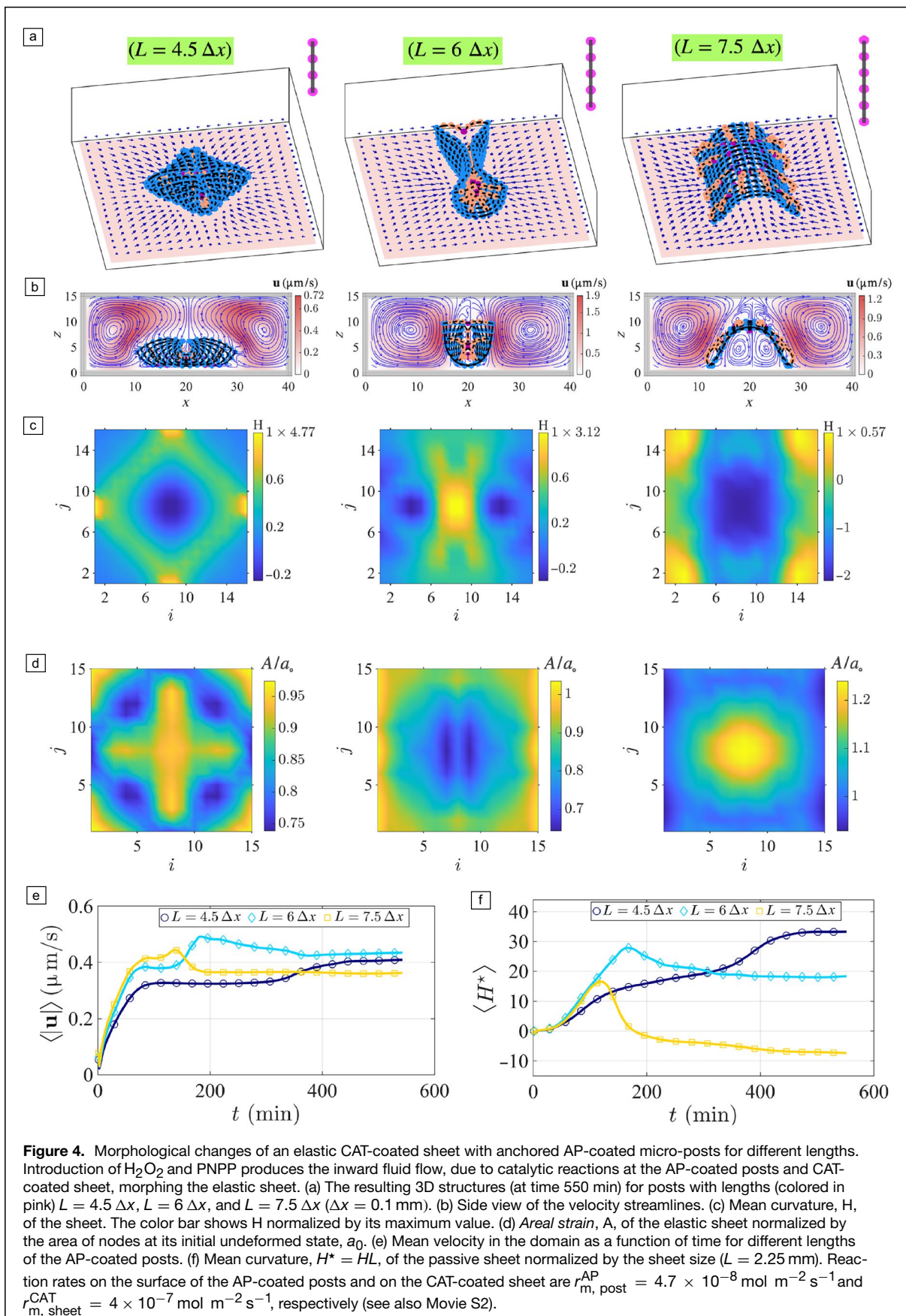


Figure 3. Morphological changes of an elastic passive sheet with anchored AP-coated micro-posts for different lengths. Introduction of PNPP produces the inward fluid flow, due to catalytic reaction at the AP-coated posts, morphing the elastic sheet. (a) The resulting 3D structures (at time 550 min) for posts with lengths (colored in pink) $L = 4.5 \Delta x$, $L = 6 \Delta x$, and $L = 7.5 \Delta x$ ($\Delta x = 0.1 \text{ mm}$). (b) Side view of the velocity streamlines. (c) Mean curvature, H , of the sheet. The color bar shows H normalized by its maximum value. (d) Areal strain, A , of the elastic sheet normalized by the area of nodes at its initial undeformed state, a_0 . (e) Mean velocity in the domain as a function of time for different lengths of the AP-coated posts. (f) Mean curvature, $H^* = HL$, of the passive sheet normalized by the sheet size ($L = 2.25 \text{ mm}$). Reaction rate on the surface of the AP-coated posts is $r_{\text{m, post}}^{\text{AP}} = 4.7 \times 10^{-8} \text{ mol m}^{-2} \text{ s}^{-1}$ (see also Movie S1).



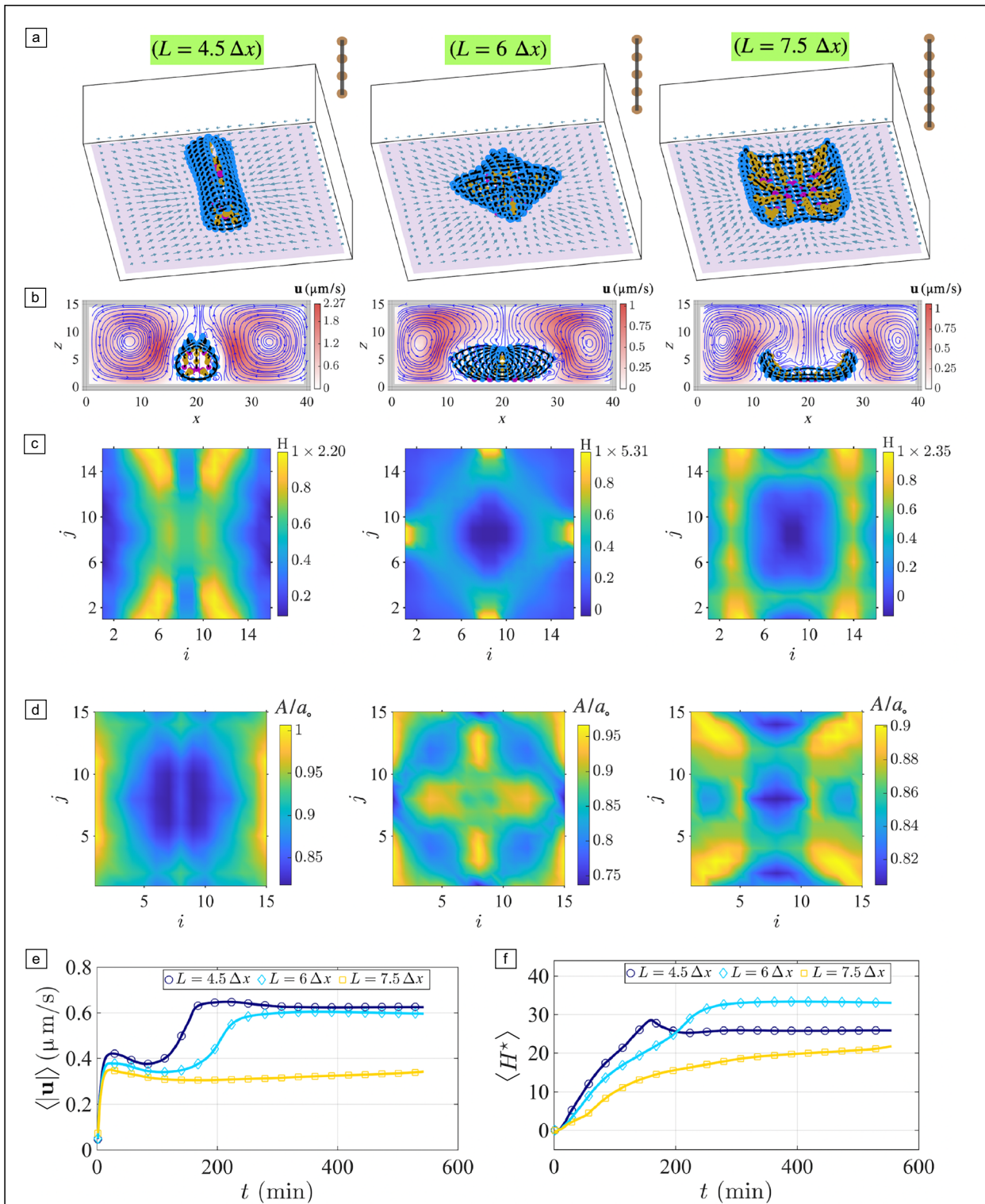
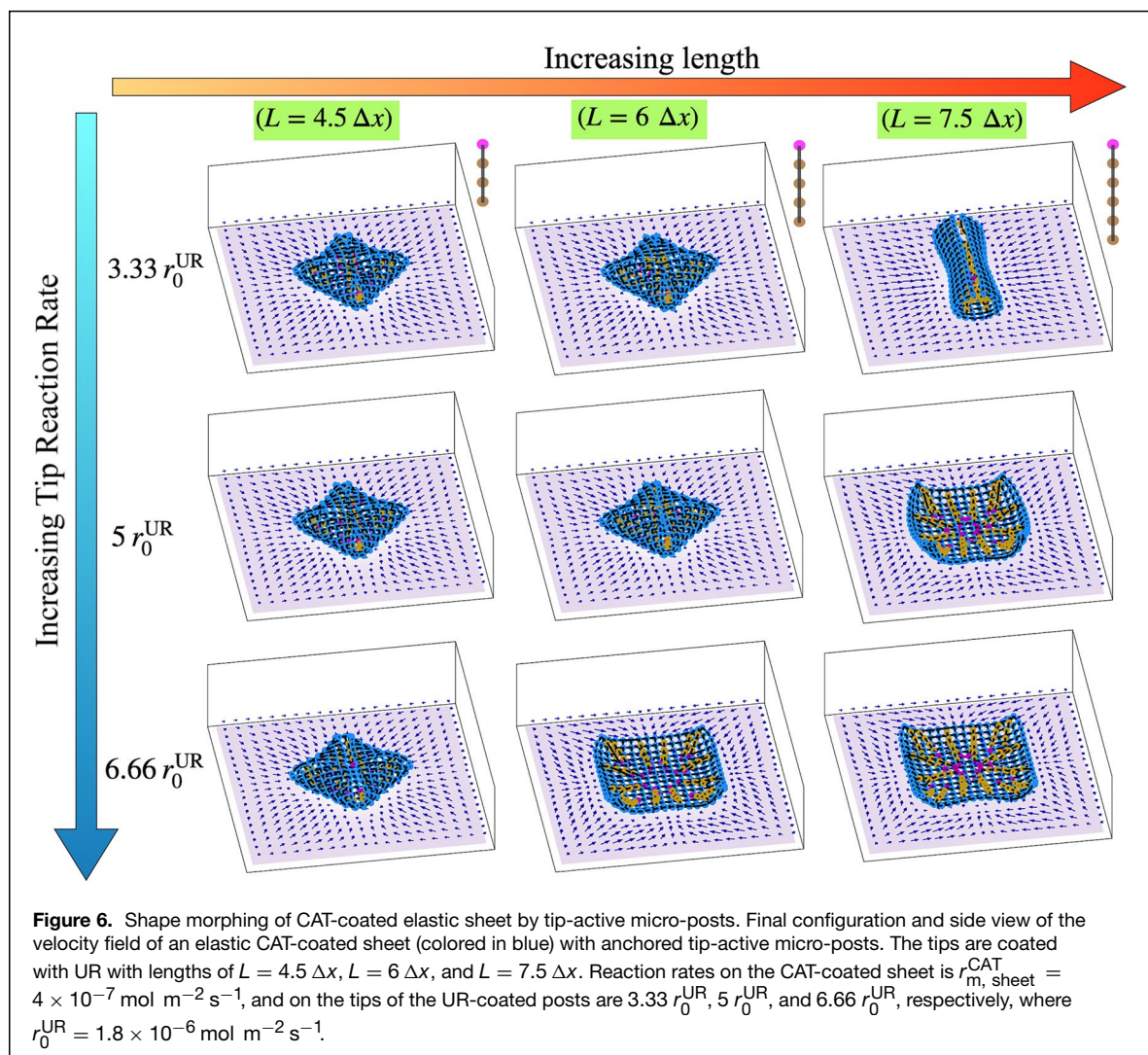


Figure 5. Morphological changes of an elastic CAT-coated sheet with anchored UR-coated micro-posts for different lengths. By introducing H_2O_2 and urea, the combination of opposing inward and outward flows generated by CAT-coated sheet and UR-coated posts generates different sheet structures for different lengths of the micro-posts. (a) The resulting 3D structures (at time 550 min) for posts with lengths (colored in brown) $L = 4.5 \Delta x$, $L = 6 \Delta x$, and $L = 7.5 \Delta x$ ($\Delta x = 0.1$ mm). (b) Side view of the velocity streamlines. (c) Mean curvature, H , of the sheet. The color bar shows H normalized by its maximum value. (d) Areal strain, A , of the elastic sheet normalized by the area of nodes at its initial undeformed state, a_0 . (e) Mean velocity in the domain as a function of time for different lengths of the AP-coated posts. (f) Mean curvature, $H^* = HL$, of the passive sheet normalized by the sheet size ($L = 2.25$ mm). Reaction rates on the surface of the UR-coated posts and on the CAT-coated sheet are $r_{\text{m, post}}^{\text{UR}} = 1.8 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1}$ and $r_{\text{m, sheet}}^{\text{CAT}} = 3.6 \times 10^{-6} \text{ mol m}^{-2} \text{ s}^{-1}$, respectively (see also Movie S3).



micro-posts that are anchored around the corners of the non-active sheet (see Movie S1 in SI Appendix). The reaction in Equation 7 creates inward convective flow, which counteracts the effects of gravity as it lifts the middle portion and edges of the sheet upward, away from the bottom wall (Figure 3a). The sheet's configuration remains stable for hours until the reactant concentration is sufficiently low so that the generated convective flow cannot deform and change the shape of the sheet.

The data in Figure 3 indicate that when the post length is $L = 0.45 \text{ mm}$, the posts produce the slowest fluid velocity (Figure 3e) and an average mean curvature that is close to zero. In comparison with the other cases in Figure 3, the chemical reactivity is not sufficient to generate sufficiently rapid flows to dramatically distort the sheets conformation.

As the length of micro-posts is increased, the corresponding increase in the number of active nodes changes the velocity profile in the domain. For micro-posts of length $L = 0.6 \text{ mm}$, the inward flow displays a higher velocity, which increases the force acting on the sides of the sheets. The resultant inward

flow pushes the edges of the sheet to yield a scroll-shaped structure with positive mean curvature. Figure 3b shows the side view of the structure with streamlines and the fluid speed at the mid-plane of the sheet (i.e., the plane $y = 2 \text{ mm}$). In this case, hydrodynamic and fluid-structure forces introduced by the chemical reactivity overwhelm the elastic forces associated with the deformation of the sheet, the steric interactions between neighboring posts and the gravitational forces pushing the nodes downward. In other words, the chemical reactivity is effective at converting chemical into mechanical energy.

The dynamic behavior of the $L = 0.75 \text{ mm}$ case illustrates the competition between the effects of chemical reactivity and mechanics as the lighter products initially transforms the flat sheet first into a scroll-like structure having positive curvature (similar to case $L = 0.6 \text{ mm}$), but at latter times, the steric interactions between sufficiently long posts on the opposing sides of the sheet ultimately lead to a bridge-like structure.

Comparison across all three structures reveals that the velocity field shows nontrivial behavior and is largest for the



micro-posts of length $L = 0.6$ mm, where the sheet has a roll-shape structure. This structure also has largest curvature compared to the structure with smaller or larger micro-posts lengths. The images in Figure 3d show as expected, that the areas exhibiting the highest curvature also exhibit the largest areal strain (as is the case in Figures 4 and 5). Taken together, the plots reveal that significant chemical reactivity (velocity) is needed to significantly distort the sheet from its flat state. Importantly, the resulting three-dimensional (3D) structures have positive (middle column), and negative (first and last column) curvatures controlled by the post length as it is shown in Figure 3c, f.

Shape morphing of an elastic CAT-coated sheet with AP-coated micro-posts

The velocity of the inward flow in the previously discussed system can be increased by coating the substrate with CAT (Figure 2b) because catalytic reactions with CAT also produce inward flow. When H_2O_2 and PNPP are simultaneously added to solution, (see Movie S2 in SI Appendix), the cumulative effects of the inward flows increase the overall fluid velocities in the domain relative to those with just AP-coated posts (see Figures 3e and 4e).

The addition of the CAT-coated substrate displays the greatest influence for the short posts with $L = 0.45$ mm. Examining similar cases shown in rows Figure 2a–b, we observe that higher reaction rates enable the flow to lift and bend the edges of the sheet inward to form an envelope-like structure (Figure 2b). In comparison with the passive sheet, the observed increase in fluid velocity (Figure 4e) provides the higher drag force needed to morph the sheet into an “envelope.” Consistent with the latter observations, the mean curvature for the $L = 0.45$ mm structure increases with time, until it lies above the forms corresponding to the longer posts (see Figures 3d and 4d).

With the increase in post height to $L = 0.6$ mm, the additional beads increase not only the net reaction rate and related magnitude of the inward fluid velocity (the speed in the lateral mid-plane is now $u = 1.9$ $\mu\text{m/s}$) but also provide a larger arc length for the fluid–structure interactions. The corresponding flow profiles in Figure 4b show that the fluids produce two symmetric vortices that are now localized around the center of the domain, sweeping the entirety of the edges upward. Consequently, the sheet is morphed into a scroll-shaped structure with positive curvature. The fact that this scroll resembles that for the passive sheet at $L = 0.6$ mm indicates that the fluid flow is generated primarily by the enzyme-coated posts, rather than the sheet.

When $L = 0.75$ mm, the increase in post length also contributes to an increase in the gravitational forces acting downward (from top to bottom) on system, pushing the edges of sheet close to the bottom wall. But, unlike the other examples in row (b) of Figure 4, the dynamic behavior at $L = 0.75$ mm is nonmonotonic. At early times, the increased chemical reactivity from the additional beads pulls the outer edges in the

y -direction upward (see Movie S2 in SI Appendix) and this structure exhibits positive curvature. (Recall that the system exhibits twofold symmetry.) Eventually, however, the structure morphs into the bridge-like shape shown in the last column in Figure 4a–b and exhibits negative curvature (see Figure 4c). In the $L = 0.75$ mm example, the posts are sufficiently long that they exhibit steric repulsion as they are all drawn to the center. This repulsion is significantly large because it pushes the clustered posts apart, causing them to point away from each other. As a result, the structure acquires a bridge-like shape. Moved away from the bottom wall, chemical reactivity generates a low velocity inward flow on the bottom side of the sheet, giving rise to the flow profile in the last panel in Figure 4b. Gravity can also be seen to play a role as the long posts do not extend freely into the solution, but lie along the nearest walls.

Notably, this bridge-like structure resembles that for the $L = 0.75$ mm case with the passive sheet. Hence, the effects of steric repulsion and gravity outweigh the effects from the sheet’s chemical reactivity for this post length.

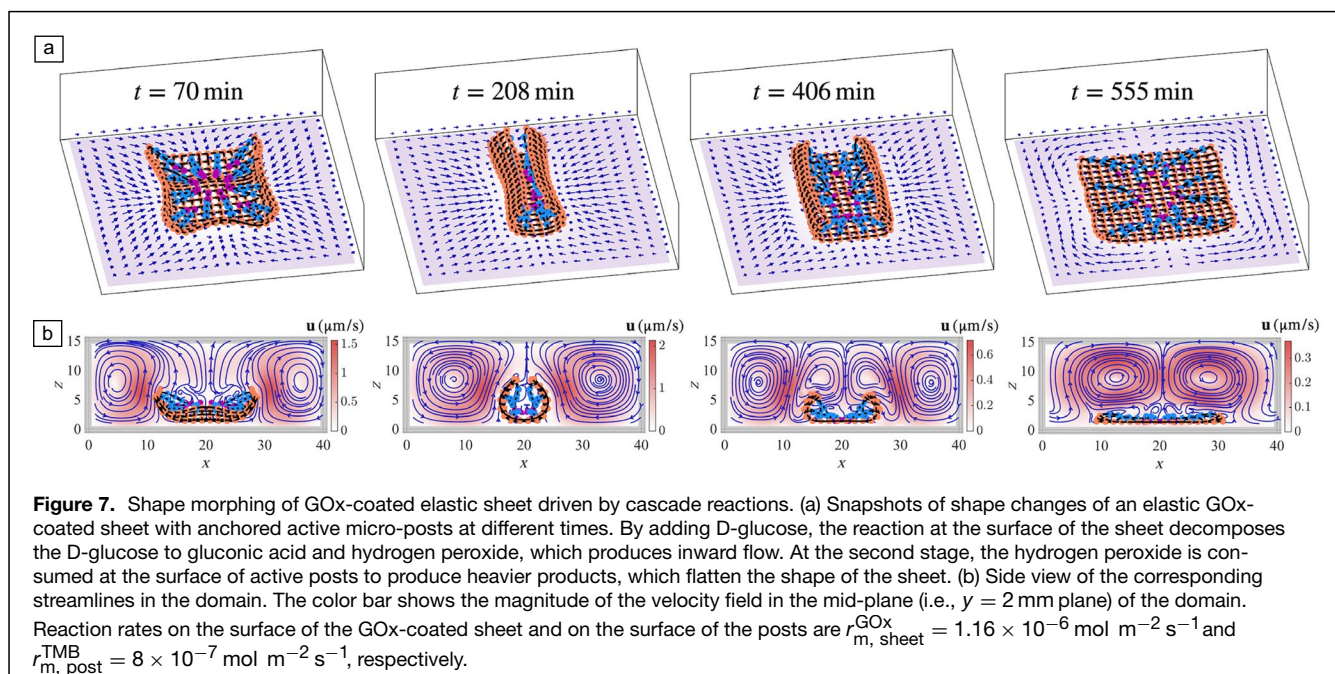
Shape morphing of an elastic CAT-coated sheet with UR-coated micro-posts

Here, we consider an active CAT-coated elastic sheet supporting UR-coated micro-posts (see Figure 5). There are two competing flows in this scenario: the reaction in Equation 6 creates inward convective flow due to lighter products generated by the active sheet, whereas the reaction in Equation 8 creates outward convective flow due to the heavier products generated by the active micro-posts (see Movie S3 in SI Appendix).

In the presence of both hydrogen peroxide and urea, two simultaneous catalytic reactions occur at the active sheet and active micro-posts. The decomposition of hydrogen peroxide to lighter products on the sheet lifts the edges of the sheet, whereas the decomposition of urea to heavier products at the micro-posts opposes the bending of the sheet and causes the sheet to retain its initial flat structure.

For the shorter lengths of the micro-posts ($L = 0.45$ mm), with less UR-coated active sites, the flow due to the CAT-coated sheet generates more inward fluid velocity to push the edges of the sheet and yield a scroll-shaped structure. As we increase the length of the micro-posts to $L = 0.6$ mm, the magnitude of the inward flow decreases (from $u = 2.27$ $\mu\text{m/s}$, to $u = 1$ $\mu\text{m/s}$), and the sheet starts to roll only on the corners to form an envelope-like structure; see Figure 5b.

By increasing the length of the active micro-posts to $L = 0.75$ mm, the related outward flow counters the inward flow produced by CAT-coated sheet (as is evident from Figure 5d). Now, the chemical reactivity introduced by the sheet is not sufficient to overcome the elastic forces associated with the sheet’s deformation and the downward gravitational force of the longer micro-posts. The mean curvature is always positive, and the largest value for the mean curvature corresponds to micro-posts with $L = 0.6$ mm, where the sheet morphs to envelope-like structure (see Figure 5e).



Functionalized tip

In the previous examples, we assumed that the posts are uniformly coated with enzymes. In these systems, the increasing post length produces two effects: it increases excluded volume interactions and raises the reaction rates associated with enzymes coating the posts. To isolate effects of the post length from the post reaction rate (proportional to the post length), we consider a case where only the tip of the posts (last bead) is coated with enzymes. In this situation, increasing the length of the posts does not affect the posts' reaction rates and we can analyze the effect of post length on the shape transformations. Simulations show results (Figure 6) in cases where the reaction rates $3.33r^{\text{UR}}$, $5.0r^{\text{UR}}$, and $6.66r^{\text{UR}}$, where r^{UR} is the reaction rate at a single urease coated post and L is set to 0.45, 0.6, and 0.75 mm.

We focus on a variant of the cases discussed directly above where the posts and sheet yield different modes of fluid motion: the enzymes (urease) on the posts' tips promote outward flow while the coating (CAT) on the sheet generates inward flow (see Movie S4 in SI Appendix). Consequently, as the reaction rate at the post tips is increased, the fluid velocity for the outward flow becomes sufficiently high so that it tends to unwrap folded sheets. As seen next, however, as the chain length is increased, the effects of steric hindrance become increasingly important and compete with effects due to chemical reactivity and hydrodynamics at the post's tip. The horizontal rows in Figure 6 show the effects due to increases in post length and steric repulsion at a fixed reactivity. The vertical columns show results from increasing the reactivity of the tip for a fixed post length.

A comparison among the different rows in Figure 6 reveals that at low chemical reactivity, steric repulsion plays a key role

when the longest posts are brought into proximity and convert an envelope into a scroll. The cases in the last row show that increasing L to 0.75 enhances the cooperative effects due to steric repulsion and the chemical reactivity producing outward flow, leading to a relatively flat sheet.

The first column in Figure 6 reveals that increasing the reactivity for the short chains ($L = 0.45$ mm) has negligible effect on the final 3D morphology. The systems fold into envelopes, where inward flow from the nearby sheet bends all corners toward the center of the sheet. Here, the chains are too short to experience steric effects. Conversely for $L = 0.75$ (last column), increasing the chemical reactivity reveals that a closed scroll is driven to unroll as the influence of chemically generated outward flow is enhanced by the effects of steric repulsion. A similar effect is seen in the second column at the highest chemical reactivity.

Cascade reactions

The chemical reactions employed in the previously discussed systems produced types of mechanical actions where the elastic layers were either "folding" or "unfolding." Once the action was accomplished and the chemical reactants were consumed, the motion could be repeated by adding fresh chemicals. On the other hand, in living systems enzymatic reactions are chained together to generate a coordinated series of movements, so that the system itself regulates a multistep process. For synthetic systems, chemical reaction cascades can be employed to produce analogous multistep functionality. In cascade reactions, the output of one reaction serves as the input for the next. The chemical specificity of catalysts plays a vital role here; the catalysts only respond to certain reagents and hence, with multiple catalysts, the initial reactant instigates the first reaction, and subsequent reactions are initiated only once the products of the previous step



are catalyzed in the solution. Hence, the system encompasses an inherent time delay between subsequent steps, which enriches the range of dynamic behavior that emerges in the chemical solution.

An example of this two-stage reaction is described by Equations 9–10.²⁷ In systems that contain enzymes glucose oxidase (GOx) and tetramethylbenzidine (TMB), the addition of glucose triggers GOx (localized on the sheet and shown with orange in **Figure 7**) to decompose glucose into hydrogen peroxide and $C_6H_{12}O_7$. The resulting hydrogen peroxide now triggers a second, subsequent step as it instigates heavier oxidation products of TMB (on the $RuTe_2$ -functionalized posts shown with blue in **Figure 7**) by decomposing H_2O_2 :



The first reaction initially curls the elastic layer up, while later the second reaction uncurls the sheet as it is shown in **Figure 7**. In particular, the first reaction generates inward fluid flows that curl an initially flat sheet (first panel on the left) into a rolled tubular structure. Once a sufficient amount of H_2O_2 is produced in the chamber, the second reaction begins to dominate by producing heavier products (TMB-ox) that generate outward flow and unroll the layer (third panel); the final effect of the second outward flow leads to a completely flattened sheet shown in panel 4 (see Movie S5 in SI Appendix).

Hence, the system shown in **Figure 7** can autonomously execute two consecutive mechanical processes. The cycle (shown in panels 1–4) can be repeated with the addition of fresh glucose. Cascades encompassing multiple such couplings (where the product of one serve as the reactant of the other) can display a more complex string of steps and thereby be designed for greater functionality, where each step permits the completion of distinct mechanical tasks.

Conclusion

While the behavior of cilia arrays in biology has been well studied,²⁸ less is known about how the movement of cilia in flowing fluids affects the morphology of the underlying, anchoring surface and how these morphological changes affect subsequent cilia–substrate interactions. It is vital to understand this interconnected behavior because of the inherent feedback in the system. Namely, the motion of the cilia due to fluid flow will exert mechanical forces on the physically connected, soft substrates, and these forces can morph the shape of the substrate. The latter morphological changes will affect the fluid–structure interactions and the fluid velocity in the system, as seen herein. But the changes in the fluid velocity will in turn affect the forces acting on the deformable surfaces and lead to feedback in fluid–structure interactions, hydrodynamics, and even the kinetics of the enzymatic reaction.²⁹ Examining the cross-talk among the fluid, the flexible cilia and the attached flexible substrate is key to furthering our understanding of

the behavior of biological cilia and developing guidelines to achieve the desired behavior in synthetic ciliated systems.

To this end, we modeled thin elastic surfaces that were decorated with an array of anchored posts and immersed in a fluid-filled chamber. Both the lateral dimensions of the substrate and the height of the posts were on the millimeter scale. Active posts were coated with various enzymes, while the substrate itself was either active or passive (noncoated). As the enzymatic reactions on the coated surfaces convert reactants into products, they generate density gradients. The resultant solutal buoyancy forces prompt the formation of circulating vortices in the chamber. If the reaction yields products that are less dense than the reactants, the vortices display inward flow, which circulates in the clockwise direction. If the products of the reaction are denser than the reactants, the circulating vortices move in the counterclockwise direction. The direction of the fluid flow can be tuned not only by introducing or altering the enzymatic coating, but also by modifying the number density, positioning, and length of the posts. Variations in these parameters enable the system to undergo controllable shape changes in the entire “carpet,” the cilia plus the substrate.

In the simplest system considered here, we focused on a passive, compliant substrate encompassing active posts coated with catalase (CAT), which reacted with the solution to generate symmetric, inward flows centered about the middle of the carpet. The flow exerted fluid drag that morphed the entire carpet into either a scroll, envelope, or a bridge-like structure. When both the substrate and posts were coated with CAT, the combined inward flows increased the fluid velocity and the effect of the drag forces. In the latter case, when the length of the posts was small, the structural transformation of the carpet was dominated by the chemical reactivity introduced by the active substrate. Distinct from the case with the passive substrate, this increase in chemical reactivity enabled the carpet to be folded into the observed envelope-like shape, where now all four corners experience an upward drag.

As the length of posts increases, it becomes progressively more difficult to enwrap the posts in an envelope or scroll-like structure. Consequently, the longer posts promote constructions such as bridges, where the cilia extend into the surrounding fluid. For longer posts, steric hindrance between these objects plays the dominant role and the geometries obtained with the CAT-coated substrate were like those obtained with the passive substrate.

We also modeled examples where the CAT-coated substrates, which produce inward flow subtended posts with a urease (UR) coating, which generated outward flow. The resulting inward and outward flows can cancel each other, but one of the flow fields can be made to play the dominate role by tailoring the ratios of the respective solutal expansion coefficients and the relative reaction rates of the active substrate and posts. For example, when the reaction rate of the substrate was fixed, increasing the post length raised the reaction rates on these sites and led to the UR prompted outward flow to dominate



over inward flow. As a result, the initial scroll-like sheets that envelop short posts were gradually unwrapped by the outward flow to form flat layers.

To isolate effects of increasing the post length from the associated increase in reaction rate, we considered cases where only the tip of the posts (last bead) was coated with enzymes. In this situation, the reaction rate of the posts does not change, and we could observe the effect of the steric hindrance between the long posts. The latter repulsive forces drove envelop-like shapes to open into scroll-like structures (Figure 6 for $3.33r^{UR}$).

To date, more experimental studies are needed to glean additional insight into the interplay between chemical reactivity and mechanics in controlling morphological changes for cilia on soft surfaces.^{28,30–32} Recent experiments by Zhang et al.³⁰ involved flexible synthetic posts tethered onto a circular inflatable surface. In contrast to our study, an external pump was used to alter the surface morphology rather than the mechanical work performed by the cilia and surrounding fluid. The experiments, however, did indicate that the morphological changes of the flexible substrate affect the global flow patterns near the ciliated surface. Other studies involving magnetically actuated cilia on soft surfaces³¹ showed that elastic cilia tethered to the underside of a soft surface could display mutual, coordinated deformations, so that the entire ciliated structure was made to crawl or roll. The latter study indicates the feasibility of fabricating functional systems that involve the coordinated, combined motion of flexible cilia and soft substrates. In particularly relevant and convincing experiments on reef coral,³² researchers showed that photocatalytic reactions on short cilia anchored to mutable soft tissue led to observable deformations of the underlying soft tissue as the reactants were converted into the products. The predictions from our computational studies and existing experiments indicate the utility of integrating the responsive behavior of elastic posts and deformable substrates for applications in soft robotics³¹ and attaining a greater understanding of biological cilia by considering the compliance of the anchoring surfaces and the mutual mechanical interactions at cilia/substrate interfaces.

The predictions from our computational studies also indicate the utility of integrating the responsive behavior of elastic posts and deformable substrates for applications in soft robotics.³¹ For example, in a fluidic environment, the sheets can act as the palm of a hand, with the posts serving as fingers. Coated with different chemical enzymes, both the palm and fingers can be manipulated independently by simultaneously introducing the appropriate reactants to instigate the motion of the respective enzyme-coated parts or adding the reactants in a time-dependent, sequential manner. Enzymatic reactions are typically biocompatible and hence would not be chemically harmful to the living organisms. In fluids, this benign process can be used to controllably open and close the “hand” (as in Figure 7), and therefore trap and extract specific particulates or species from the solution. In the latter case, the sheet and pillars are sufficiently soft that they could maintain the viability

of the trapped living species, which can be removed for future studies. In effect, the hands would mimic a functionality of biological cilia on living organisms, where the motion of the cilia effectively aids in extracting the organism’s “food” from the surrounding solution.

Author contributions

A.C.B. designed the system described here. M.M. and O.E.S. performed the simulations and numerical calculations. All the authors helped analyze the data and contributed to writing the manuscript.

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Data availability

Data produced for this manuscript are available upon reasonable request to the authors.

Conflict of interest

There are no conflicts to declare.

Supplementary information

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